

THE OPEN DOOR: INSTANT CHATTING WITH WEBSOCKETS

WEBSOCKETS — REAL-TIME COMMUNICATION FOR KIDS

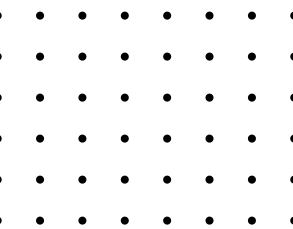


WAITING FOR UPDATES?



THE CRICKET SCORE STORY

Imagine you're at home, eagerly waiting for the latest IPL cricket score. You keep refreshing your phone again and again, hoping for an update. "Did Virat hit a six? Did we win?" Every few seconds, you tap refresh... but why does it take so long? How do we get scores so fast?



TRADITIONAL POLLING

Imagine asking your friend the same question again and again: "Any news yet?" That's how traditional polling works! Apps keep asking the server "Is there an update?" over and over. This method is slow and uses more battery on your phone.

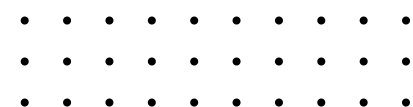


SLOW!

THE OPEN DOOR

WHAT ARE WEBSOCKETS?

- Imagine having a door open all the time to talk instantly! WebSockets keep a connection open between your device and the server, so messages can flow back and forth without waiting.
- Unlike knocking on a door and waiting for someone to answer each time, WebSockets are like having a friend's door always open — you can chat instantly whenever you want!

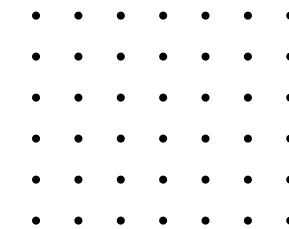
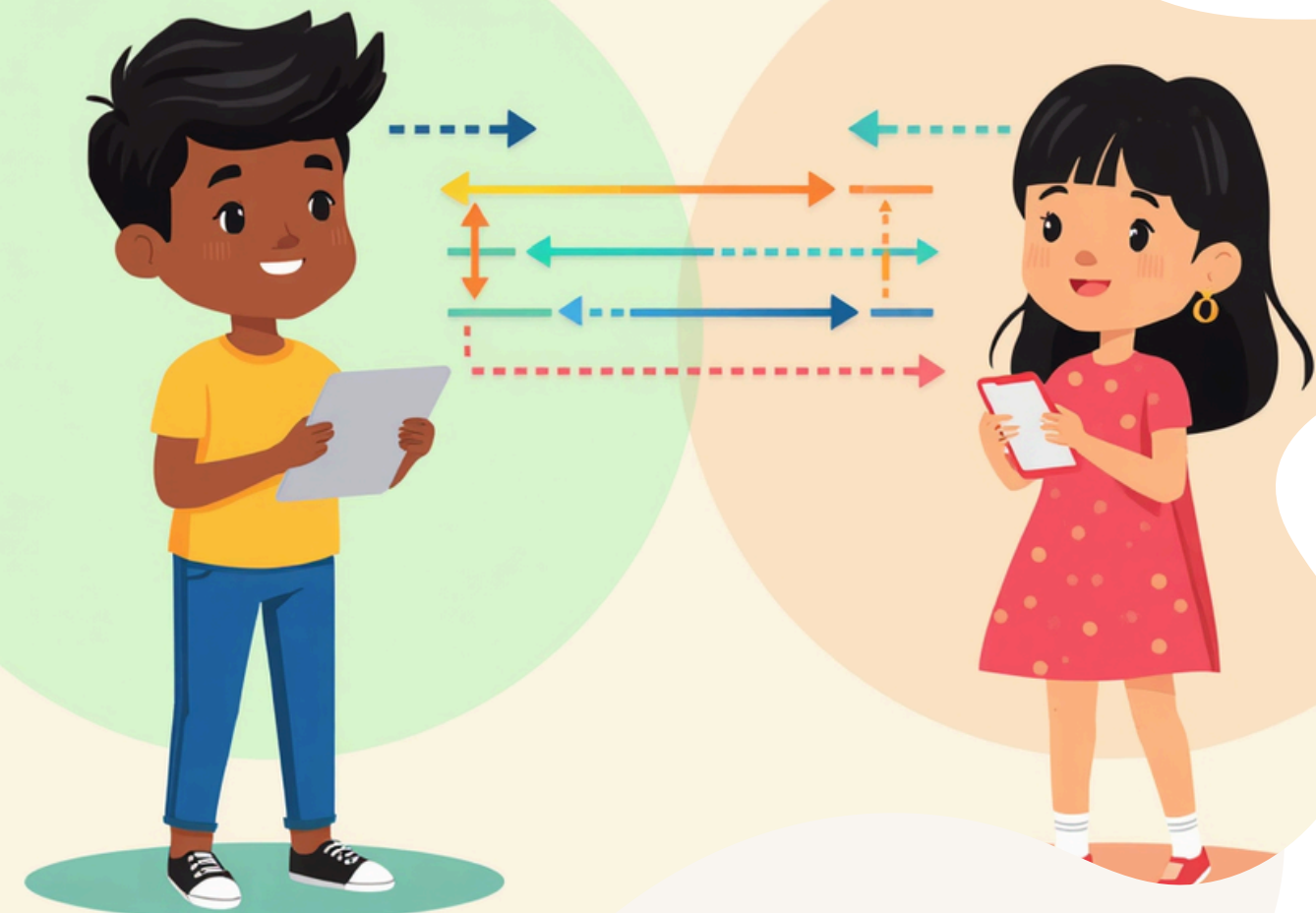


TALKING BOTH WAYS

BI-DIRECTIONAL COMMUNICATION



- With WebSockets, messages can flow both ways anytime without waiting! It's like having a two-way walkie-talkie that's always on. You send a message, and your friend receives it instantly. They reply, and you get it right away too — no delays, no waiting!



REAL-WORLD EXAMPLES

> WHERE WE USE WEBSOCKETS

- Live cricket score updates during IPL matches — see runs and wickets instantly!
- Instant messaging apps like WhatsApp let you chat in real-time
- Online games use WebSockets for smooth multiplayer action



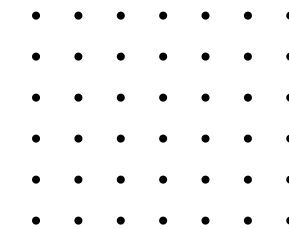
HOW POLLING AND WEBSOCKETS DIFFER



Polling: Your app keeps asking "Any updates?" again and again. It's slower and uses more battery — like knocking on a door every few seconds!



WebSockets: The door stays open! Messages flow instantly both ways. It's faster and saves battery — like having a walkie-talkie always on!



INTERACTIVE ACTIVITY

GUESS THE DELAY!

Time to test what you've learned! Think about how polling and WebSockets work. One keeps asking "Any updates?" while the other keeps the door open. Which method do you think delivers messages faster?



RAISE YOUR HAND TO VOTE!



DISCUSS WITH YOUR PARTNER



HOW WEBSOCKETS WORK

SIMPLE STEPS

1. Client opens a WebSocket connection — like knocking on a door
2. The door stays open — no need to knock again!
3. Messages travel instantly both ways — you talk, your friend replies right away

It's that simple! Once connected, chatting happens in real-time.

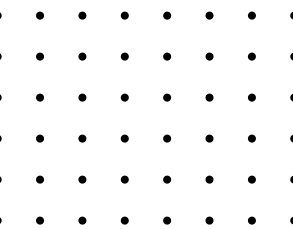


BENEFITS OF WEBSOCKETS

- ☑ Instant updates in real-time — no waiting for cricket scores or chat messages!
- ☑ Saves battery and data on your phone, making apps run smoothly and efficiently.



100%
SMOOTH



INTERACTIVE QUIZ

TRUE OR FALSE?

Test what you learned! Read each statement and decide if it's TRUE or FALSE. Ready to show off your WebSockets knowledge?



"WebSockets keep the door open all the time." — TRUE ✓



"POLLING ASKS THE SERVER AGAIN AND AGAIN." — TRUE ✓

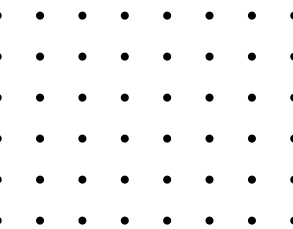


WHY WEBSOCKETS ARE COOL



LET'S REMEMBER WHAT WE LEARNED!

- Traditional polling is slow and wastes battery by asking again and again
- WebSockets keep the connection open like an always-open door
- Messages travel instantly both ways — no waiting!
- Used in apps and games you love: WhatsApp, IPL scores, online games

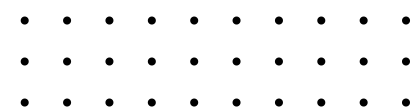


STAY TUNED!

EXPLORE MORE!

YOUR CODING ADVENTURE AWAITS!

- Ready to become a WebSocket wizard? In Session 2, we'll learn how to build our very own instant chat application!
- You'll discover how real apps like WhatsApp and online games use WebSockets to make chatting super fast and fun!



THANK YOU!

THANKS FOR JOINING! ANY QUESTIONS?

